

Disparities in Cervical Cancer Screening Among Women With and Without a History of Cancer: Evidence from BRFSS 2024

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Cervical cancer screening remains high but significant disparities across education, race/ethnicity, and healthcare access. While cancer survivors engage with healthcare more frequently, it is unclear whether this translates to higher cervical screening adherence. Using 2024 BRFSS data, this study examined screening disparities among 88,157 women aged 21–65 and found that healthcare access and not survivorship was the dominant driver of adherence. Women with lower education had 18% lower screening, racial/ethnic minorities 3–12% lower, and those without a regular provider up to 24% lower. Both survivors and non-survivors fall short of the Healthy People 2030 target of 84%, highlighting the need for structural interventions to reduce screening inequities.

Background

- About 25% of eligible U.S. women are not up to date with cervical screening, with persistent disparities by race/ethnicity, education, and healthcare access.
- 17+ million cancer survivors in the U.S. have more frequent healthcare contact, yet cervical screening adherence among survivors often falls below national targets, unlike breast and colorectal screening.
- Healthcare access, having a provider and recent checkout is a known driver of screening disparities, but limited research examines whether these access-driven disparities differ for cancer survivors vs. the general population.
- Understanding these patterns is critical for developing equitable, targeted screening interventions.

Research Goals

- Examine disparities in cervical screening prevalence between women with and without cancer history.
- Identify structural drivers (healthcare access, education, race/ethnicity) of screening gaps.
- Assess whether these disparities differ by age group (21–39 vs. 40–65).

Methods

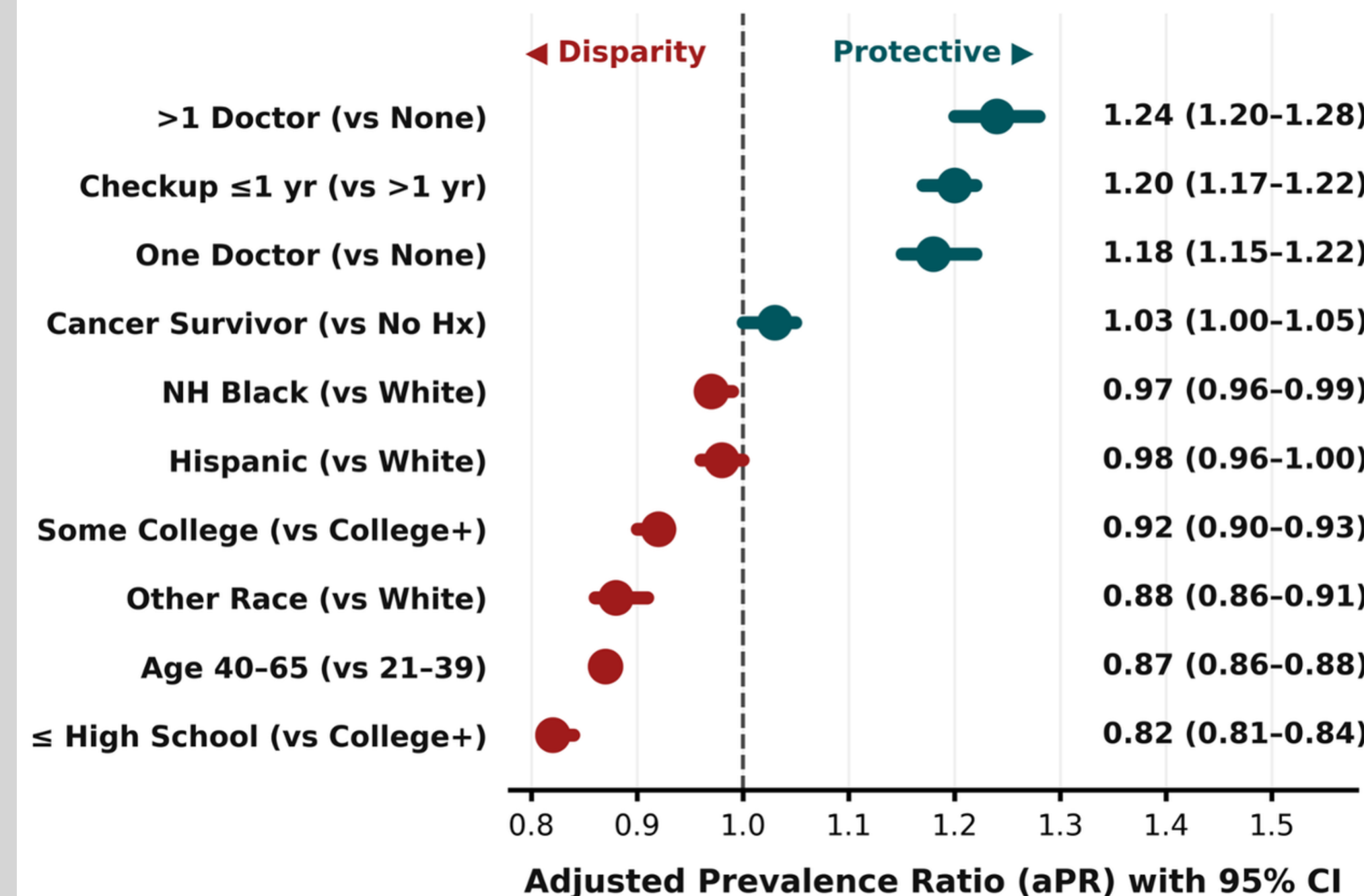
- Data: 2024 BRFSS, nationally representative survey with complex sampling design.
- Population: Women aged 21–65; excluded skin cancer only and missing data. Final N = 88,157.
- Outcome: Up-to-date cervical screening (Pap ≤3 yrs or HPV ≤5 yrs) from _PAPHPV1.
- Exposure: Non-skin cancer history (CHCOCNC1 = 1).
- Covariates: Age, race/ethnicity, education, recent checkout, personal doctor.
- Analysis: Survey-weighted Quasi-Poisson regression; progressive model building (4 models); aPRs with 95% CIs; effect modification by age.
- Software: R v4.5.1, survey package.

Measures and Results

- N = 88,157 women; 4,665 (5.3%) cancer survivors. Survivors were older (88% aged 40–65 vs. 64%) with greater healthcare access (88% recent checkout vs. 81%).
- Screening: 80.9% (survivors) vs. 78.4% (non-survivors).
- Unadjusted: PR = 1.03 → + Demographics: 1.05 → + Education: 1.05 → + Healthcare Access: 1.03*
- Top predictors: Checkout ↑20%, Doctor ↑18–24%. Disparities: ≤HS ↓18%, Age 40–65 ↓13%, minorities ↓3–12%.

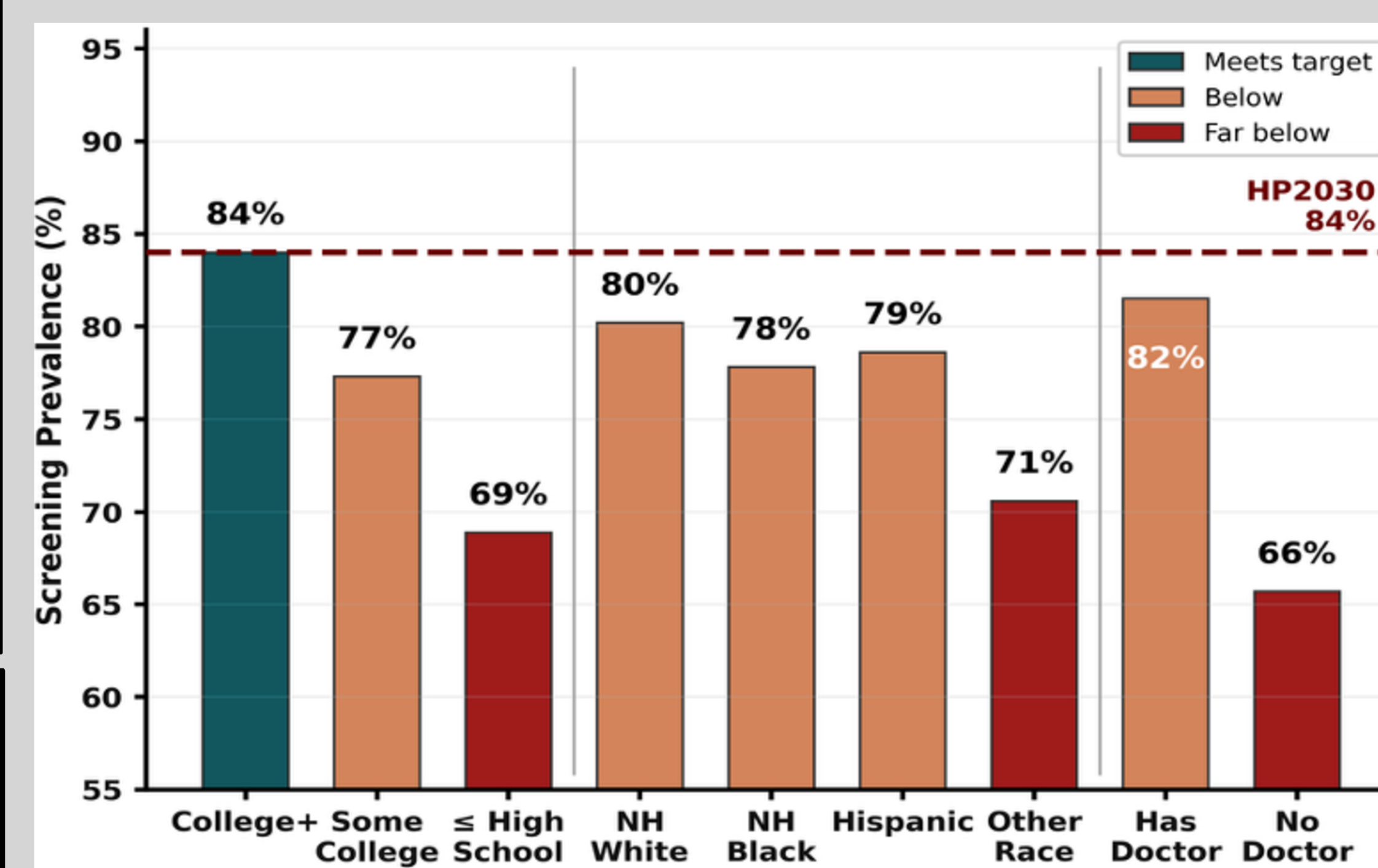
Measures and Results, continued

Fig 1. Predictors of Cervical Screening Adherence (Fully Adjusted Model)



Key Finding: Healthcare access (↑20–24%) was the strongest predictor of screening, far outweighing cancer survivorship (↑3%).

Fig 2. Screening Prevalence by Subgroup vs. Healthy People 2030 Target (84%)



Conclusions & Community Relevance

- Despite survivors marginal screening advantage (aPR = 1.03), both survivors (80.9%) and non-survivors (78.4%) fall below the Healthy People 2030 target of 84%, revealing a system-wide screening gap.
- Stark disparities persist: women with ≤high school education had 18% lower screening, racial/ethnic minorities 3–12% lower, and women aged 40–65 had 13% lower adherence.
- Healthcare access was the single strongest driver of screening, recent checkout (↑20%) and having a personal doctor (↑18–24%) far outweighed all other predictors, highlighting access as a root cause of disparities.
- These findings underscore that closing screening gaps requires addressing structural barriers to care, not just cancer awareness, particularly for women with lower education, racial/ethnic minorities, and those lacking a regular provider.
- Future research should examine screening disparities by specific cancer type and time since diagnosis, and develop targeted interventions for underserved communities.

Discussion

- Progressive modeling revealed negative confounding, survivors' older age masked the true effect; adjusting for demographics strengthened the association to 5%.
- After adding healthcare access, the effect returned to 3%, confirming access drives survivors' screening advantage.
- Healthcare access (↑20–24%) far outweighed survivorship (↑3%), indicating screening gaps are structurally driven.
- Interventions should prioritize expanding access to providers for underserved populations.

Contact Information

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Note: The work presented here was done for the purposes of a course and is not my thesis or dissertation.

Screening gaps are driven by disparities in healthcare access, education, and race/ethnicity, closing these gaps requires structural solutions, not just awareness.